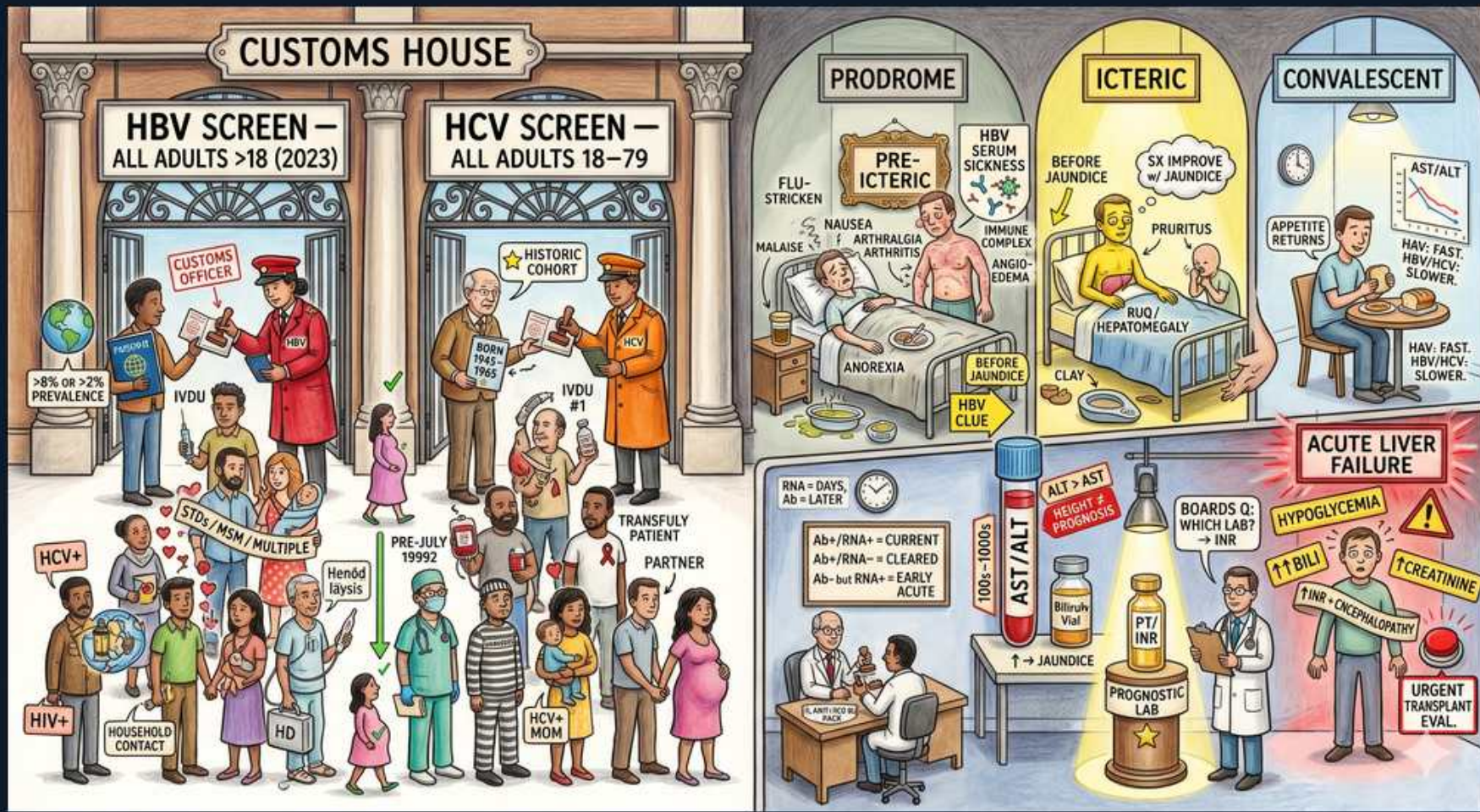


Viral Hepatitis — Acute Clinical Course & ALF



1. Prodrome / pre-icteric

WHAT YOU SEE

PRODROME bay header with PRE-ICTERIC bed

WHAT IT ENCODES

Prodromal (pre-icteric) phase: 1-2 weeks of flu-like malaise, fatigue, anorexia, nausea/vomiting, low-grade fever, and RUQ discomfort before any jaundice appears. Viral shedding and infectivity (especially HAV) are often HIGHEST during this phase, before the patient even looks ill.

BOARD TRAP

Constitutional symptoms classically PEAK in the prodrome and then improve as jaundice appears — a stem saying the patient feels worst once jaundiced is the trap. Also, a patient is most contagious before jaundice, not after.

2. Serum-sickness prodrome (HBV)

WHAT YOU SEE

'HBV SERUM SICKNESS' arthralgia placard in prodrome bay

WHAT IT ENCODES

HBV can produce a serum-sickness-like prodrome from circulating immune complexes: symmetric arthralgias/arthritis, urticaria or maculopapular rash, and occasionally fever and proteinuria. Other immune-complex phenomena tied to HBV include polyarteritis nodosa (PAN) and membranous glomerulonephritis.

BOARD TRAP

An arthralgia + urticaria prodrome before hepatitis points to HBV specifically (immune complexes), not HAV. If the stem also describes necrotizing vasculitis/renal involvement, think HBV-associated polyarteritis nodosa.

3. Anorexia / aversion

WHAT YOU SEE

ANOREXIA label in prodrome bay

WHAT IT ENCODES

Profound anorexia, nausea, and the classic distaste/aversion to cigarettes and fatty foods characterize the prodrome of acute viral hepatitis. These nonspecific symptoms often precede any lab or jaundice clue.

BOARD TRAP

Nonspecific anorexia/nausea is easy to attribute to gastroenteritis or pregnancy — in the right exposure context (travel, IDU, new partner) check hepatitis labs rather than dismissing it.

4. Icteric phase — jaundice

WHAT YOU SEE

ICTERIC bay header with jaundiced patient

WHAT IT ENCODES

Icteric phase: clinically apparent jaundice/scleral icterus, dark (tea-colored) urine from conjugated bilirubinuria, pale clay-colored stools, pruritus, and tender hepatomegaly. Jaundice becomes visible once total bilirubin exceeds ~2.5-3 mg/dL.

BOARD TRAP

Many adults (and most acute HCV) are ANICTERIC — absence of jaundice does not exclude acute hepatitis. And constitutional symptoms usually improve once jaundice begins, not worsen.

5. Pruritus / cholestasis

WHAT YOU SEE

PRURITUS placard, icteric bay

WHAT IT ENCODES

Cholestatic features — pruritus, clay stools, dark urine, and conjugated (direct) hyperbilirubinemia — dominate the icteric phase. A prolonged cholestatic variant of HAV can last weeks with marked itching but still resolves fully.

BOARD TRAP

Pruritus + conjugated hyperbilirubinemia can mimic biliary obstruction; in acute viral hepatitis the ducts are normal on ultrasound, so don't order ERCP — image first to exclude stones, then recognize the hepatocellular pattern.

6. Convalescent recovery

WHAT YOU SEE

CONVALESCENT bay, patient eating, appetite returns

WHAT IT ENCODES

Convalescent phase: appetite returns, jaundice fades, and aminotransferases trend back toward normal over weeks. HAV recovers fastest and completely; HBV is slower; HCV often smolders into chronic infection rather than truly convalescing.

BOARD TRAP

Recovery speed and chronicity diverge by virus: HAV fast/complete, HBV slower (~5% adult chronicity), HCV mostly chronic. Don't assume normalizing ALT means cure in HCV — confirm with HCV RNA.

7. ALT/AST >1000

WHAT YOU SEE

Lab bench 'ALT/AST ↑1000s' test tube

WHAT IT ENCODES

Acute viral hepatitis causes marked aminotransferase elevation, frequently into the hundreds-to-thousands, with ALT > AST (hepatocellular pattern). The degree of transaminitis does NOT correlate with prognosis — synthetic function (INR) does.

BOARD TRAP

AST:ALT ratio >2:1 with only modestly elevated values (typically <300-400) suggests ALCOHOLIC hepatitis, not viral. Very high ALT > AST = viral/toxic/ischemic. Don't read 'AST > ALT' as viral hepatitis.

8. Bilirubin ↑ jaundice

WHAT YOU SEE

'↑JAUNDICE' bilirubin tube on lab bench

WHAT IT ENCODES

Total (and conjugated) bilirubin rises during the icteric phase, producing jaundice, dark urine, and pale stools. Bilirubin level reflects disease activity but not severity — a high bilirubin alone does not mean liver failure.

BOARD TRAP

Don't use jaundice/bilirubin level to gauge severity or decide transplant — a deeply jaundiced patient with normal INR and no encephalopathy is not in liver failure. Severity is judged by INR and mental status.

9. PT/INR is the key prognostic

WHAT YOU SEE

'PT/INR' prognostic lab placard

WHAT IT ENCODES

Prothrombin time / INR is the single most important prognostic marker in acute hepatitis because clotting factors (II, VII, IX, X) have short half-lives and reflect real-time hepatic synthetic function. A rising INR signals impending liver failure. Factor VII falls earliest.

BOARD TRAP

The dangerous pattern is FALLING transaminases with RISING INR and worsening mental status — this means hepatocytes are dying out (less enzyme to release), not recovery. Don't be reassured by a dropping ALT if the INR is climbing.

10. ALF danger room

WHAT YOU SEE

Red 'ACUTE LIVER FAILURE' danger room, right

WHAT IT ENCODES

Acute liver failure (ALF) = hepatic ENCEPHALOPATHY + COAGULOPATHY (INR ≥ 1.5) developing within ~26 weeks (often days-weeks) in a patient WITHOUT pre-existing chronic liver disease. Leading causes: acetaminophen (#1 in US), viral (HAV/HBV±HDV/HEV), drug reactions, ischemia.

BOARD TRAP

Encephalopathy is REQUIRED to call it ALF — coagulopathy alone (acute hepatitis with high INR but normal mentation) is not ALF. Also, pre-existing cirrhosis disqualifies the 'acute liver failure' label (that's acute-on-chronic).

11. Hypoglycemia

WHAT YOU SEE

HYPOGLYCEMIA warning in ALF room

WHAT IT ENCODES

Hypoglycemia is a key danger sign in ALF because the failing liver loses gluconeogenesis and glycogen stores — check glucose frequently and give IV dextrose. Other ALF complications: cerebral edema/raised ICP, infection, AKI/hepatorenal, lactic acidosis.

BOARD TRAP

In an encephalopathic ALF patient, don't assume worsening mental status is purely hepatic — rule out reversible HYPOGLYCEMIA (and infection) first. Treat glucose before escalating.

12. Rising bili + creatinine

WHAT YOU SEE

'↑T.BILI' and '↑CREATININE' placards in ALF room

WHAT IT ENCODES

Worsening total bilirubin, rising creatinine (hepatorenal syndrome/AKI), and deepening encephalopathy together signal poor prognosis and trigger urgent transplant evaluation. These map onto King's College Criteria (e.g., for non-acetaminophen ALF: INR >6.5 , or any 3 of age, etiology, jaundice-to-encephalopathy interval, bilirubin, INR).

BOARD TRAP

Transplant referral is driven by validated criteria (King's College / INR + creatinine + encephalopathy + etiology), NOT by how sick the patient subjectively looks or how high the transaminases are.

13. Urgent transplant eval

WHAT YOU SEE

'URGENT TRANSPLANT EVAL' red placard, ALF room

WHAT IT ENCODES

ALF management = ICU-level supportive care (glucose, treat cerebral edema, manage coagulopathy/infection) PLUS early referral to a transplant center, because survival without transplant in severe cases is poor. For acetaminophen-induced ALF, give N-acetylcysteine; NAC may also benefit non-acetaminophen ALF.

BOARD TRAP

Do NOT wait for the patient to 'declare' or for transaminases to fall — refer to a transplant center as soon as ALF criteria are met. Delayed referral is the wrong-answer trap; also don't forget NAC even when the cause isn't clearly acetaminophen.